# Lumbar Spine MRI

**ACG: A-0059 (AC)**
**Link to Codes**

**MCG Health**
Ambulatory Care
29th Edition

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## Clinical Indications

- Lumbar spine MRI may be indicated for **1 or more** of the following(1)(2)(3)(4)(5):
  - Ankylosing spondylitis or other seronegative spondyloarthritis,[A] suspected, and plain x-ray results nondiagnostic(21)(22)(23)(24)(25)(26)
  - Cancer or neoplasm evaluation, staging, or surveillance needed, as indicated by **1 or more** of the following(34)(35)(36)(37)(38)(39):
    - Bone neoplasm (benign or malignant) involving spine, known or suspected (eg, chondrosarcoma, chordoma, Ewing sarcoma family of tumors, giant cell tumor of bone, osteosarcoma, metastatic tumors), as indicated by **1 or more** of the following(47)(48)(49):
      - Initial evaluation or staging(50)
      - Monitoring disease progression or response to treatment
      - Post-treatment surveillance
    - Histiocytic or dendritic cell neoplasm[B] with involvement of vertebrae, known or suspected(51)(54)(55)
    - Localized lumbar back pain and **1 or more** of the following(56)(57):
      - Bone scan positive(50)(58)
      - Pain occurs mainly at night.(45)(59)
      - Persistent back pain in patient older than 50 years(60)
      - Personal history or concurrent diagnosis of malignancy(40)(41)(42)(61)(62)
      - Weakness, rapidly progressing(60)
      - Weight loss, unexplained(60)
    - Myelopathy signs or symptoms (eg, motor weakness, bowel or bladder dysfunction)(40)(41)(42)(63)(64)
    - Primary central nervous system tumor (eg, adult-type or pediatric-type diffuse glioma, ependymoma, medulloblastoma, other central nervous system embryonal tumor, primary central nervous system lymphoma, meningioma), and need for **1 or more** of the following(48)(65)(66)(67)(68)(69):
      - Brain tumor, and need for evaluation for spinal involvement
      - Initial evaluation, staging, or treatment planning
      - Monitoring disease progression or response to treatment
      - Post-treatment surveillance
    - Primary non-central nervous system tumor, and need for diagnosis, staging, or monitoring of known or suspected spinal cord, spine, or leptomeningeal metastasis, as indicated by **1 or more** of the following(48)(49)(65)(68)(69)(70):
      - Known spinal cord, spine, or leptomeningeal metastasis, and need for monitoring disease progression or response to treatment

        - Post-treatment surveillance
        - Suspected spinal cord, spine, or leptomeningeal metastasis (eg, due to myelopathy, pain, finding on other imaging), and need for evaluation, staging, or treatment planning
    - Screening for central nervous system tumor in patient with high-risk genetic disorder or syndrome (eg, neurofibromatosis type 2, von Hippel-Lindau syndrome)(69)(71)(72)(73)(74)
- Cerebrospinal fluid leak, suspected(34)(75)(76)

Congenital spinal conditions, suspected, as indicated by **1 or more** of the following(34)(77)(78)(79):
- Anterior sacral meningocele(80)
- Caudal regression syndrome(81)
- Diastematomyelia (ie, split cord malformation)(80)
- Dorsal dermal sinus(81)
- Hemangioma(82)(83)(84)
- Hydromyelia
- Intradural lipoma
- Myelocele
- Myelomeningocele(80)
- Split notochord syndrome(81)
- Tight filum terminale(81)

Infection, known or suspected (eg, vertebral osteomyelitis, disk space infection, epidural abscess), as indicated by **ALL** of the following(4)(34)(45)(85)(86)(87):
- Risk factors for spinal infection, as indicated by **1 or more** of the following:
    - Bone scan or plain x-ray suggestive for infection(94)
    - C-reactive protein or erythrocyte sedimentation rate elevated(95)(96)(97)
    - Fever(98)(99)
    - History of intravenous drug abuse, other source of infection, or recent invasive procedure(100)
    - Immunosuppression(101)
    - Positive blood culture(95)(102)
    - Recent history of spinal surgery(103)
    - Tuberculosis, concurrent or suspected(93)(101)(104)(105)
- Suspected spinal infection (eg, localized midline back pain, limp, refusal to walk)(95)(98)

Inflammatory or demyelinating process, suspected (eg, clinically isolated syndrome, multiple sclerosis, other demyelinating disease), as indicated by **1 or more** of the following(34)(106)(107):
- Multiple sclerosis, suspected, as indicated by **1 or more** of the following(108)(109)(112)(118):
    - Ascending numbness or tingling (eg, from foot to trunk)(111)
    - Brown-Sequard syndrome(111)
    - Conditions mimicking multiple sclerosis (eg, Sjogren syndrome, systemic lupus erythematosus, antiphospholipid syndrome) cannot yet be excluded.(64)(119)(120)
    - MRI of brain nondiagnostic for multiple sclerosis(121)
    - Signs or symptoms of myelopathy or myelitis(99)(122)(123)
- Transverse myelitis in spinal cord, suspected (idiopathic or in conjunction with multiple sclerosis), as indicated by **1 or more** of the following(101)(124)(125)(126):
    - Bilateral signs or symptoms of involvement of appropriate level of spinal cord
    - Clearly defined sensory level
    - Sudden onset of sensory, motor, and autonomic dysfunction attributable to appropriate level of spinal cord

Pain localized to low back or radicular in nature, subacute or chronic, as indicated by **1 or more** of the following(57)(58)(127)(128)(129):
- Adult and **ALL** of the following:
    - Failure to improve after 6 or more weeks of nonoperative treatment, as indicated by **1 or more** of the following(4)(58)(99)(136)(137)(138)(139):
        - Activity modification(129)
        - Exercise(140)
        - Pharmacotherapy (eg, analgesics, anti-inflammatory medications)(141)(142)
        - Physical therapy(142)(143)(144)
    - Patient being considered for invasive treatment (eg, epidural steroids, surgery)(136)(145)
    - Significant interference with daily function
- Child or adolescent and **ALL** of the following(45):
    - Age younger than 18 years
    - Concern for serious underlying pathology (eg, malignancy, infection, inflammatory process), as indicated by **1 or more** of the following:
        - Associated signs suggestive of cancer (eg, easy bruising, abdominal mass, lymphadenopathy)
        - Fever
        - Gait abnormalities or refusal to walk or crawl

        - Morning stiffness
        - Nocturnal pain or constant pain
        - Neurologic deficit
        - Pain lasting more than 4 weeks or worsening pain
        - Unintentional weight loss
- Postoperative spinal complications, known or suspected, as indicated by **1 or more** of the following(146)(147):
    - Differentiating recurrent disk herniation from epidural fibrosis, soft tissue inflammation, facet joint inflammation, or bone marrow edema
    - Postoperative hemorrhage or hematoma(103)
    - Pseudomeningocele(148)
    - Spondylodiscitis(149)
    - Sterile arachnoiditis(150)
- Scoliosis, as indicated by **1 or more** of the following(34)(151)(152):
    - Congenital scoliosis
    - Early-onset scoliosis (age 9 years or younger)
    - Neurofibromatosis
    - Presurgical planning for adolescent idiopathic scoliosis to assess possible neural axis malformation, as indicated by **1 or more** of the following(151)(155)(156)(157):
        - Abnormal neurologic findings on clinical examination(158)
        - Age at first visit 10 years or younger
        - Kyphosis at curve apex
        - Left-sided thoracic curvature
        - Male gender
        - Pain, moderate to severe(6)(158)
        - Rapid curve progression (ie, more than 1 degree per month)
        - Short segment curve (ie, less than 6 vertebral segments)
        - Thoracic kyphosis 30 degrees or greater
        - Vertebral abnormalities (eg, hemivertebrae, block vertebrae) detected on x-ray
- Spinal cord compression, myelopathy, or cauda equina syndrome, suspected (eg, positive Babinski sign, depressed ankle deep tendon reflexes, gait abnormality, hyperreflexia, sensory or motor deficits)(34)(42)(58)(101)(159)(160)(161)(162)(163)
- Spinal cord infarction or ischemia, suspected (eg, sudden-onset back pain, sensory or motor deficits, urinary retention)(164)
- Spinal cord stimulator placement planning needed, and no MRI of region within 12 months prior to placement(169)(170)
- Spinal dysraphism, suspected, as indicated by **1 or more** of the following(78)(80)(174)(175)(176)(177)(178):
    - For infant, as indicated by **1 or more** of the following(180)(181)(182)(183):
        - Absent reflexes
        - Aplasia cutis congenita
        - Decreased spontaneous leg movement
        - Decreased urinary stream
        - Deviation of gluteal cleft
        - Foot asymmetry(81)
        - Leg atrophy
        - Midline lumbosacral skin lesions (eg, atypical dimple, dermal sinus, lipoma, tail)(6)(81)
        - Unilateral lumbosacral hypertrichosis
    - For toddler: developmental delay in walking, abnormal gait
    - For older child: spasticity, hyperreflexia, bowel or bladder disturbance, including incontinence(81)
- Spinal stenosis of lumbar spine, suspected, as indicated by **ALL** of the following(34)(184)(185)(186):
    - Neurogenic claudication with progressive or disabling symptoms, as indicated by **1 or more** of the following(187)(188)(189)(190)(191):
        - Back, leg, or buttock pain worsens with prolonged standing and activities requiring lumbar extension.(185)
        - Bilateral leg or buttock pain with prolonged walking that is relieved with sitting or leaning forward(129)
        - Leg weakness(192)
        - Wide-based gait or abnormal Romberg test(138)(193)
    - Patient being considered for invasive treatment(194)
- Spondylolysis, known or suspected by radiologic evidence (eg, by plain x-ray, bone scan, CT scan), with or without spondylolisthesis, and **1 or more** of the following(44)(195)(196)(197)(198):
    - Focal neurologic findings
    - Low back pain that worsens with activity
    - Urinary retention or incontinence(200)
- Stereotactic spine radiotherapy treatment planning(201)(202)(203)(204)(205)
- Syringomyelia in lumbar spine, suspected, as indicated by **1 or more** of the following(34)(208)(209)(210):
    - Muscle wasting in appropriate lumbar spine dermatomes
    - Sensory loss in appropriate lumbar spine dermatomes

- Weakness in appropriate lumbar spine dermatomes

- Tethered cord, suspected, as indicated by **1 or more** of the following(80)(174)(176)(180)(211)(212):
  - Anorectal malformation(181)(213)(214)
  - Cutaneous manifestations of occult spina bifida (eg, nevus, lipoma, tufts of hair, hemangioma, dimple overlying spine, asymmetric gluteal cleft, dermal sinus tract)(181)(182)(215)
  - Gait abnormality or difficulty
  - Urinary dribbling or lack of bladder control(81)(98)
  - Urodynamic tests abnormal
- Trauma or fracture of lumbar spine, known or suspected, as indicated by **1 or more** of the following(216)(217)(218)(219):
  - Epidural hematoma, suspected(227)(231)
  - Fall from height greater than 10 feet (3.1 meters)(232)
  - Malignancy, known or suspected(58)(230)(233)(234)
  - Myelopathy(235)(236)
  - Neurologic findings that correspond to vertebral level above level of injury seen on plain x-ray or CT scan(236)(237)
  - Neurologic symptoms associated with traumatic mechanism of injury, as indicated by **1 or more** of the following(34)(232)(238)(239)(240)(241):
    - Ejection from motorcycle(242)(243)
    - High-speed motor vehicle accident (greater than 50 mph (80.5 kph))(242)(243)
    - Intoxicated state(244)
    - Minimal trauma in osteoporotic patient with suspected fragility fracture
    - Motor vehicle accident, with acute hyperflexion of spine over seat belt(245)
  - Osteoporosis, known or suspected(58)(230)(232)(246)
  - Pediatric patient(222)(247)(248)(249)(250)
  - Plain x-ray or CT scan results nondiagnostic(220)(221)(229)
  - Pre-existing spinal pathology(232)(251)
  - Progressive neurologic deficit(235)
  - Radiculopathy(237)
- Tuberculosis of spine, known or suspected, as indicated by **1 or more** of the following(93)(252)(253):
  - Diagnostic evaluation of suspected spinal tuberculosis
  - Preparation for surgical management
- Repeat evaluation of specific area or structure with same imaging modality, and **ALL** of the following:
  - Clinical need for repeat imaging, as indicated by **1 or more** of the following:
    - Change in clinical status (eg, worsening symptoms or new associated symptoms), and findings may impact treatment
    - Need for interval reassessment that may impact treatment plan
    - Need for re-imaging either prior to or after performance of invasive procedure
  - Prior imaging results of specific area or structure with same imaging modality documented and available for comparison

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# Evidence Summary

## Background

Some gadolinium-based contrast agents used for diagnostic and treatment guidance with MRI or MRA have been associated with nephrogenic systemic fibrosis in patients with chronic kidney disease or acute kidney injury.(6)(7)(8)(9)(10) **(EG 2)** Gadolinium deposition in the brain has been reported by multiple research groups; however, the clinical significance of this finding is unclear.(6)(7)(11)(12) **(EG 2)** Specialty society guidelines recommend weighing the risks vs benefits when using gadolinium-based contrast agents for MRI.(13) **(EG 2)** MRI-based techniques may be contraindicated in patients with ferromagnetic intracranial aneurysm clips and certain types of ferromagnetic foreign bodies and cochlear implants.(14) **(EG 2)** Although these implants have traditionally included cardiac pacemakers and implantable cardioverter-defibrillators, multiple trials have validated the safe and effective use of pacemaker systems designed for MRI use, and a guideline supported by multiple specialty societies states that many patients with implantable cardiac devices, including those not originally intended for MRI scanning, can safely undergo MRI scanning under defined clinical indications.(7)(15)(16)(17)(18) **(EG 1)**

## Criteria

The evidence for the clinical indications found in this guideline includes 206 published peer reviewed articles, 16 specialty society or other evidence-based guidelines, and 12 book sections.

For ankylosing spondylitis or other seronegative spondyloarthritis, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Review articles recommend that MRI of the spine, including the sacroiliac joints, be reserved for patients with inflammatory back pain when plain x-ray is nondiagnostic.(27)(28)(29)(30)(31) **(EG 2)** A cross-sectional study of 50 children with juvenile idiopathic arthritis found that 62% had abnormalities such as vertebral compression fracture; spinal canal narrowing; or disk degeneration, protrusion, or collapse noted on thoracic and lumbar spine MRI.(32) **(EG 2)** An international multispecialty clinical practice guideline on the diagnosis and management of spondyloarthritis recommends conventional radiography for patients with suspected axial disease but notes that MRI of the sacroiliac joints is an alternative in younger patients and those with a short duration of symptoms. Sacroiliac MRI is also recommended for patients with nondiagnostic x-rays.(33) **(EG 2)**

For cancer or neoplasm evaluation and staging, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Spine MRI is the preferred imaging modality for suspected metastatic disease to the spine in patients with known primary malignancy; metastatic epidural spinal cord compression is seen in 5% to 10% of patients with cancer.(40)(41)(42) **(EG 2)** Spinal cord tumor in children may present with persistent localized back pain that is worse at night.(43)(44)(45) **(EG 2)** Additional findings in children with spinal cord tumor may include burning pain with paresthesia, gait disturbance, paraspinal muscle spasm, progressive motor weakness, progressive scoliosis, and sensory deficits.(46) **(EG 2)**

For cerebrospinal fluid leak, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline notes that MRI of the spine may be useful to discover the location of a cerebrospinal fluid leak in patients with intracranial hypotension.(34) **(EG 2)**

For congenital spinal conditions, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline indicates that MRI is an appropriate imaging modality for assessment of congenital spinal conditions.(34) **(EG 2)**

For infection of the spine, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRI is a preferred option for diagnosis of suspected spinal infection and inflammation, including spondylitis, epidural abscess, osteomyelitis, and spinal tuberculosis.(88)(89)(90)(91)(92) **(EG 2)** Because spinal tuberculosis can present in noncontiguous vertebral bodies, patients with known spinal tuberculosis should undergo MRI imaging of the entire spine.(93) **(EG 2)**

For inflammatory or demyelinating processes, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Diagnostic criteria have been developed for inflammatory or demyelinating processes, such as clinically isolated syndrome or multiple sclerosis (most commonly manifested as unilateral optic neuritis), brainstem syndrome, or partial myelitis, based on the demonstration of central nervous system lesions disseminated in space and disseminated in time on MRI imaging of the brain and, if appropriate, on spinal cord imaging.(108)(109)(110)(111)(112)(113) **(EG 2)** Spinal cord involvement is seen on MRI imaging in 80% to 90% of patients with multiple sclerosis; asymptomatic spinal cord lesions may be present in up to 40% of patients with clinically isolated syndrome, with lesions occurring more commonly in the cervical spine than in the thoracic spine.(114)(115)(116)(117) **(EG 2)** An expert consensus panel recommends spinal cord imaging for patients with suspected demyelinating disease who present with signs or symptoms of spinal cord involvement or if the number of lesions on brain MRI is insufficient to meet established criteria for multiple sclerosis.(108) **(EG 2)**

For pain localized to the low back or radicular in nature, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Systematic reviews found no evidence that early lumbar spine imaging with MRI made a difference in clinical outcome in patients without serious underlying conditions, such as cancer, infection, or cauda equina syndrome.(130)(131) **(EG 1)** A prospective cohort study of 5239 patients age 65 years or older compared early imaging, including either CT scan or lumbar MRI, with delayed or no imaging and found no differences in clinical outcomes at 12 months.(132) **(EG 2)** In the absence of urgent ("red flag") indications, such as suspected infection, cancer, significant neurologic compromise, or inflammatory disease, expert consensus guidelines and review articles note that diagnostic imaging is usually not clinically helpful in the early course of an episode of back pain.(4)(58)(133)(134)(135) **(EG 2)** For children and adolescents with back or neck pain, a specialty society guideline notes that, while imaging is not generally indicated for pain that is self-limited or not associated with signs or symptoms of an underlying disorder, MRI is appropriate for evaluating pain associated with potential red flags (eg, associated gait abnormalities, unintentional weight loss, persistence over more than 4 weeks).(45) **(EG 2)**

For postoperative spinal complications, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles indicate that MRI is effective at demonstrating common complications after spinal surgery.(146)(148)(149) **(EG 2)**

For scoliosis, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Review articles conclude that MRI is not routinely indicated for pain in the absence of abnormal neurologic examination findings or an atypical curve, such as a left-sided curve (levoscoliosis), a short segment, or kyphosis near the curve apex.(153)(154)(155)(156) **(EG 2)** However, MRI is useful preoperatively in evaluating the neural axis.(157) **(EG 2)**

For spinal cord compression, myelopathy, or cauda equina syndrome, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Ominous signs such as urinary or fecal incontinence, urinary retention, lower extremity motor weakness, or gait abnormalities warrant urgent MRI of the spine, which is the preferred imaging modality for detection and confirmation of malignant spinal cord compression.(63) **(EG 2)** Literature reviews of acute transverse myelitis recommend urgent neuroimaging with MRI to rule out cord compression.(107)(124) **(EG 2)**

For spinal cord infarction or ischemia, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** MRI of the spine is the preferred diagnostic procedure to detect spinal cord ischemia. The clinical manifestations may evolve over minutes to hours, vary depending on the vascular territory involved, may be bilateral or unilateral, and may present transiently (also called spinal transient ischemic attacks).(164)(165)(166) **(EG 2)** Etiologies may vary (eg, aortic dissection, atherosclerosis, coagulopathy, embolization, mechanical compression, systemic hypoperfusion, vasculitis).(164)(166)(167)(168) **(EG 2)** For patients experiencing spinal infarct or ischemia, timely identification facilitates reperfusion intervention, which may improve outcomes.(164)(165) **(EG 2)**

For spinal cord stimulator placement planning, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Preoperative MRI of the spine is recommended within 12 months prior to the spinal cord stimulator placement to evaluate potential pathology that could alter the diagnosis or negatively impact the outcome and to assess the need for alternative approaches to placement of the stimulator electrodes.(171)(172)(173) **(EG 2)**

For spinal dysraphism, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** Spinal dysraphism includes a variety of malformations with incomplete fusion of midline mesenchymal, bony, and neurologic structures; a broad bony defect of the vertebral arch is seen at the level of the malformation.(78)(80)(175) **(EG 2)** Spinal dysraphism may be associated with neurogenic scoliosis. Intraspinal pathology has been reported in 12% to 50% of patients with congenital scoliosis; the MRI findings may include anomalies such as filum lipoma, syrinx, or Chiari malformation.(154)(179) **(EG 2)**

For spinal stenosis of the lumbar spine (with or without spondylolisthesis), evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Specialty society guidelines and review articles recommend MRI of the lumbosacral spine as the preferred imaging test.(184)(185)(186) **(EG 2)**

For spondylolysis, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRI is the imaging modality of choice for investigation of lumbar spondylolysis, especially in younger patients, because it is most sensitive to early and subtle changes such as bone marrow edema and avoids exposure to ionizing radiation.(196)(197)(198) **(EG 2)** A specialty society guideline states that CT scan is the most reliable test to diagnose a pars interarticularis defect; however, lumbar spine MRI is recommended to identify neuroforaminal stenosis in patients with spondylolisthesis.(199) **(EG 2)**

For stereotactic spine radiotherapy, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** CT scan and MRI are increasingly used for treatment planning prior to stereotactic spine radiotherapy for spinal metastases.(201)(202)(204)(206)(207) **(EG 2)**

For syringomyelia, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** A specialty society guideline indicates that MRI is an appropriate imaging modality for assessment of syringomyelia.(34) **(EG 2)** A review article notes that when Chiari malformation is suspected on cranial CT scan, MRI of the brain and the whole spine is indicated.(210) **(EG 2)**

For tethered cord, evidence demonstrates a net benefit, but of less than moderate certainty, and may consist of a consensus opinion of experts, case studies, and common standard care. **(RG A2)** MRI is considered to be the study of choice when evaluating tethered cord, as it demonstrates the level of the conus and nature of the filum, identifies the cause of the tethering, and provides anatomic details needed for preoperative planning.(174)(180) **(EG 2)** A retrospective study of 790 patients with anorectal malformation reported that the incidence of tethered cord was 36%, and noted that screening for tethered cord using MRI or ultrasound has become standard practice for newborns with anorectal malformations.(213) **(EG 2)**

For trauma or fracture of the lumbar spine, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** Systematic reviews of studies of patients with spinal cord injuries who did not have abnormalities on normal plain x-rays or CT scans demonstrated abnormal spine MRI findings in 57% of children and 93% of adults.(220)(221) **(EG 1)** In a retrospective cohort study of 206 pediatric patients evaluated for spinal injuries in a trauma center, MRI was obtained more frequently in patients between 0 and 3 years of age (67%) as compared with patients between 4 and 9 years of age (20%). The authors hypothesized that this disparity may be due to difficulty in obtaining reliable neurologic examinations in infants and toddlers. MRI is also a better tool for imaging cartilage, of which a significant portion of the spinal column in this age group is composed.(222) **(EG 2)** A study of patients with 152 thoracic or lumbar vertebral compression fractures showed that specific combinations of MRI findings were able to distinguish malignant, osteoporotic, and infective etiologies with accuracy ranging from 76% to 98%, with biopsy as the gold standard.(223) **(EG 2)** Review articles note that MRI is helpful in the assessment of acute and subacute posttraumatic spinal cord injury, including damage to disks and spinal cord ligaments, as well as the assessment of spinal canal stenosis, edema, and hematoma.(224)(225)(226)(227) **(EG 2)** MRI may also be helpful at predicting higher-risk compression fractures that result in nonunion.(228) **(EG 2)** A specialty society practice guideline stated that MRI may be used to evaluate the acute thoracic and thoracolumbar traumatic spine to assess for soft tissue injury, detect fractures, and ascertain the need for surgery. The authors noted that a small study population and heterogeneity limited conclusions about the use of MRI in this setting; further long-term studies with a higher level of evidence were recommended.(229) **(EG 2)** Another specialty society guideline recommended that MRI be performed in patients with osteoporotic or malignancy-related vertebral compression fractures who are being considered for vertebral augmentation (ie, vertebroplasty, kyphoplasty) to help determine the age and healing status of the fracture.(230) **(EG 2)**

For tuberculosis of the spine, evidence demonstrates at least moderate certainty of at least moderate net benefit. **(RG A1)** MRI is considered the most accurate imaging modality for diagnosis and classification of tuberculosis, including early spinal cord changes.(93)(254) **(EG 2)**

## Rationale

Use of this MCG care guideline helps the clinician determine if a particular treatment, medication, or service might be appropriate for a specific patient, taking into account their unique health complexities.

Use of these evidence-based clinical criteria to support decision making benefits the patient by identifying patient-specific complex clinical factors and conditions, promoting personalized treatment. Utilizing evidence-based clinical criteria promotes patient safety by helping ensure that potential patient benefits outweigh the risks. In addition, the use of evidence-based guidelines can increase consistency in treatment thresholds, leading to less variation in care and promoting equitable treatment among patients.

## Related CMS Coverage Guidance

This guideline supplements but does not replace, modify, or supersede existing Medicare regulations or applicable National Coverage Determinations (NCDs) or Local Coverage Determinations (LCDs).

**Code of Federal Regulations (CFR):** 42 CFR 419.22(255); 42 CFR 422.101(256)

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**Medicare Coverage Determinations:** Medicare Coverage Database(260)

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# Footnotes

[A] Seronegative spondyloarthropathies include psoriatic arthritis, reactive arthritis (formerly Reiter syndrome), enteropathic-related arthritis, and undifferentiated spondyloarthritis.(19)(20) [ A in Context Link 1 ]

[B] Histiocytic and dendritic cell neoplasms are a heterogeneous group of rare hematologic disorders that arise from neoplastic proliferations of cells in the monocytic, histiocytic, or dendritic lineage that arise from common myeloid progenitors.(51)(52) They are classified on the basis of clinical, radiographic, histologic, and molecular features. Major categories include the plasmacytoid dendritic cell neoplasms; Langerhans cell and other dendritic cell neoplasms, including Langerhans cell histiocytosis; and histiocytic neoplasms, including Erdheim-Chester disease and Rosai-Dorfman disease.(51)(52)(53) Depending on the subtype, histiocytic and dendritic cell neoplasms may be characterized by vertebral lesions.(51)(54)(55) [ B in Context Link 1 ]

# Codes

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